Chapter 4

Producing Long-Form Content

IN THIS CHAPTER

- **Composing scholarly articles with the help of GenAI**
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This chapter is all about the role of GenAI in the production of long-form content. The focus is primarily on the capabilities of GenAI to handle large data sets and generate long-form text that's coherent, factually accurate, and contextually appropriate. The discussion also extends to the practicalities of a writer's interaction with GenAI, evaluating the division of labor in which the technology is responsible for data processing and initial text generation, and the human writer is tasked with guiding the narrative and ensuring the content's uniqueness.

Long-form content is by far the most difficult to produce successfully and to professional standards with GenAI. If you come into this thinking that GenAI is an easy way out of arduous tasks, you're gonna be sorely disappointed. However, many real benefits exist to using AI in this way and,

with practice and experience, you will get better at using it in your writing process. I promise it's worth it in the end!

Writing Academic Papers with GenAI Assistance

Tackling the creation of an academic paper can be a complex task, and GenAI is emerging as a valuable tool in this intellectual endeavor. This section provides a clear overview of how GenAI can assist in the research and writing process while maintaining the integrity and originality that academic work demands. You explore practical ways in which GenAI can serve as an aid in organizing literature, synthesizing data, and generating drafts, all without compromising your or another scholar's voice and analytical perspective.



REMEMBER The use of GenAI in academic writing should be approached with a judicious blend of skepticism and openness — recognizing its potential to streamline the research process while being wary of over-reliance on its outputs. Focus on integrating GenAI's capabilities with critical thinking and ensure that the final paper reflects a well-argued, original contribution to the field.

Using GenAI for research and academic writing

When it comes to academic writing, thorough research is the backbone that supports the entire structure of your scholarly work. GenAI is becoming an increasingly valuable tool in the researcher's toolkit and the overall AI stack, offering a suite of functions that can aid in the meticulous task of gathering and analyzing data. Consider the following:



- **TIP** **GenAI has the potential to uncover obscure connections between data points.** By analyzing vast networks of academic papers, GenAI can suggest relationships and patterns that might not be immediately obvious to human researchers. This can lead to novel insights and interdisciplinary connections that could be the seed for groundbreaking research. (However, it's important to ap-

proach these AI-generated connections with a discerning eye, validating them through further research and critical evaluation.)

- **Before you dive deep into a topic, GenAI can quickly scan and summarize existing literature on the subject.** This broad overview can help you to identify key themes, gaps in the research, and seminal works. The high-altitude reconnaissance can save valuable time and help shape the direction of your inquiry.
- GenAI can also assist in the more granular aspects of research, such as data mining and extraction. For instance, it can sift through large datasets to extract relevant statistics, trends, and figures that are pertinent to your research question. Additionally, GenAI's language-processing capabilities can be harnessed to transcribe and translate materials, making non-native language sources more accessible and broadening the scope of your research.

The use of GenAI in this domain requires a strategic approach to ensure that it serves as an enhancement rather than a replacement for the researcher's own critical-thinking skills. The following sections help guide you through process and product options (although, please know that these are examples and not your only options).

Examining GenAI tool options for research and academic writing

GenAI has evolved to include specialized tools tailored for academic research, enhancing the researcher's ability to delve into complex topics with increasing precision and depth. For instance, tools like Iris.ai (`iris.ai`) and Semantic Scholar (`www.semanticscholar.org`) use AI to map out scientific literature and provide researchers with relevant papers, reviews, and patents, often uncovering material that traditional search engines might miss. These platforms use natural language processing to understand the context of research queries, ensuring that the results are not just keyword-based but semantically related to the research question.

Another example is Zotero (`www.zotero.org`), a free reference manager that helps manage and cite research sources for book or report writing and can be integrated with GenAI tools. Zotero can automatically extract citation details from PDFs and web pages and add such works to a research folder with a simple click on a browser extension, making it easier for researchers to build and maintain their bibliographies.

You can easily connect Zotero with Research Rabbit (www.researchrabbit.ai) to further streamline your research and get your reports and papers found by other researchers, too. Research Rabbit is a free research tool built to help you discover and organize relevant papers for your research. It provides interactive visualizations, collections, digests, and data search tools. If you want a more in-depth look at Zotero and its many capabilities, you might want to view "How to use Zotero's full potential [The AI Revolution in Zotero]" on YouTube at https://www.youtube.com/watch?v=gA3o2MlnPBQ .

You can add ChatGPT to Zotero in a few easy steps. Then, you can research and write your academic paper or book in one tool. The easiest integration to add GenAI to Zotero is ARIA, an acronym for AI Research Assistant and a Zotero plugin. Before using ARIA, you need to have an OpenAI API key, which is a security code for developers to use that enables computer programs to interact. You can get an OpenAI API key here: https://help.openai.com/en/articles/4936850-where-do-i-find-my-openai-api-key . Follow the in-app instructions to add the key, select the ChatGPT model you want to use, and restart Zotero.

The integration process is super easy and takes only a few steps. After you've obtained an OpenAI API key, take the following steps to install ARIA as an add-on tool for Zotero:

1. **Go to Aria on GitHub at** https://github.com/lifan0127/ai-research-assistant .
2. **Scroll down to the Installation section and download the latest release (.xpi file) from GitHub:** https://github.com/lifan0127/ai-research-assistant/releases/latest
3. **Go to Zotero and select Tools from the top menu bar. Click on Addons.**
4. **On the Add-ons Manager panel, click the gear icon at the top right corner and select Install Add-on From File.**
5. **Select the .xpi file you just downloaded and click Open, which will start the installation process.**
   You're done!



TIP For a more detailed look at this installation process, see the walk-through tweet by Mushtaq Bilal, PhD, Syddansk Universitet on Twitter (now X) at `https://x.com/MushtaqBilalPhD/status/1735221900584865904`.

Additionally, platforms like BenchSci (`www.benchsci.com`) apply AI to decode complex scientific figures, allowing researchers to find specific experimental data without combing through countless articles manually. However, you'll need to integrate this, too, with a GenAI tool trained for writing, or copy its outputs to a GenAI text generator like ChatGPT or Claude so you can write your article or paper. You can also find specialized text generators, like SciWriter (`www.sciwriter.ai`), but many of those have waiting lists and lean more toward writing articles than writing papers.

Refining GenAI research outputs

GenAI outputs "as is" will almost never be what you want and need them to be. Indeed, the process of trying to get GenAI to produce exactly what you need can be exhausting and frustrating. Also, GenAI, even the specialized text generators, do a lousy job of writing anything in long form. It's smarter and more productive to resign yourself to these unfortunate facts and develop your writing process accordingly. In other words, focus on GenAI's strengths in data discovery and analysis and in helping you fill in any blanks in your work or thinking. Otherwise, accept GenAI's flaws and use your own mind and talent to work around them.

To refine AI research outputs and better ensure their relevance and accuracy, follow this step-by-step process:

1. **Start with a clear and concise research question or set of keywords.** Input these into the GenAI tool as a prompt to generate an initial list of resources.
2. **Use a specialized GenAI or other tool's filtering options to narrow down the results to the most relevant papers.**

This may include sorting by date, relevance, citation count, or journal impact factor. You can integrate GenAI with research tools, as discussed above, to do this work, or you can prompt a GenAI tool to do this type of sorting for you. You may want to try both ways and compare the results.

3. **Skim through the abstracts provided by the AI to assess the pertinence of each source.**
   At this stage, look for the research's main objective, methodology, and conclusions to determine its applicability to your work.
4. **For sources that pass the abstract review, read the full text to evaluate the quality and depth of the research.**
   Pay particular attention to the study's design, the robustness of the data, and the soundness of the arguments presented.
5. **Check the references and citations within the papers to find additional material and to verify the AI's suggestions.**
   This step helps to ensure that key sources haven't been overlooked.
6. **Perform a manual search for literature on your topic to compare against the AI-generated list.**
   This can help identify any gaps or biases in the AI's output.
7. **Use the AI to help extract and organize key data points, statistics, and quotes from the selected sources.**
   Ensure that all extracted information is accurately represented and properly cited.
8. **Critically evaluate the synthesized information, looking for logical consistency, potential biases, and the strength of the evidence.**
9. **Refine your search query based on insights gained from the literature and repeat the process as necessary to uncover additional resources or to narrow the focus of your research.**



REMEMBER By integrating these advanced GenAI tools into your research process and rigorously refining their outputs, you can enhance the breadth and depth of your academic writing. It's crucial to remain actively engaged in each step, using your expertise to guide the AI and to critically assess its contributions to your scholarly work.

Finding hidden pitfalls or AI limitations

While GenAI tools can offer significant advantages in academic research, researchers should be aware of some hidden pitfalls and limitations.

Understanding these limitations is essential to knowing when to supplement or replace AI research with manual efforts and other types of tools.

- **Data quality and bias:** GenAI tools are only as good as the data they have been trained on. If the underlying data contains biases or inaccuracies, the AI's outputs will reflect those flaws. Researchers should be cautious of over-reliance on AI-generated data without cross-referencing it against other sources.
- **Scope of training:** Some AI tools may have been trained on specific types of literature or within particular disciplines, which can limit their ability to identify relevant research outside of those areas. Researchers should verify whether the AI's scope aligns with their field of study and supplement with manual searches and other types of tools as needed.
- **Contextual misunderstandings:** AI may struggle with understanding the nuanced context of certain research queries, which can lead to irrelevant or tangential results. Researchers should be prepared to interpret the results critically and refine their queries to improve relevance.
- **Overlooked novel research:** AI tools may favor sources with more citations or those that are more easily accessible, potentially overlooking newer, less-cited yet highly relevant research. Researchers should manually keep abreast of the latest publications in their field to capture cutting-edge studies.
- **Non-textual data limitations:** GenAI is predominantly text-based and may not effectively analyze or interpret non-textual data such as complex graphs, images, or raw datasets. Researchers should manually review such materials to ensure comprehensive understanding and integration into their work.
- **Updates and maintenance:** AI databases require regular updates to include the latest research. If an AI tool is not well-maintained, it may miss recent publications. Researchers should check the update frequency of AI tools and supplement with manual or other tool searches in fast-moving fields.
- **Citation context:** GenAI can provide citations, but it may not always capture the context in which a study is cited. Researchers should manually review citations to understand how a source is being used within the literature to avoid misinterpretation.
- **Language limitations:** While some AI tools offer translation capabilities, nuances, and subtleties in language can be lost. For research in languages other than those the AI was primarily designed for, researchers should verify translations and interpretations by different means.
- **Depth of analysis:** AI can help identify broad patterns and connections, but it may not capture the depth of analysis required for a thorough literature review. Researchers should supplement AI findings with deep, manual analysis to uncover subtle arguments and theoretical frameworks.

To mitigate these pitfalls, maintain a balanced approach, using GenAI as a starting point for identifying potentially relevant research and then applying your own expertise and critical-thinking skills to conduct a more nuanced and comprehensive review.



WARNING Whatever you do, do not hand your boss or publish any AI-based research or written report without thoroughly reviewing and vetting first. If you do, you're likely gambling your entire career, and you're probably going to lose badly.

Supporting the academic publication process

The journey from research to publication is often a marathon filled with hurdles, from formatting to peer review. GenAI is emerging as a useful resource in this process, offering capabilities that can help streamline various stages of academic publishing. However, while GenAI can assist with some aspects of the academic publishing process, like literature reviews, writing, and organization, the more nuanced human elements involving expert judgment, context, and domain knowledge remain challenging for AI to fully automate or replicate at this stage. You should be very cautious about overestimating GenAI's role.

While GenAI tools have the potential to assist with various tasks, some of the capabilities mentioned or touted by enthusiasts and vendors — and even in GenAI responses — may not be fully developed or widely available. Here's an overview of where things stand as of this writing:

- **Manuscript formatting:** While there are tools like LaTeX that can help with formatting academic papers and reference management software like Zotero and EndNote that can format citations according to journal styles, claims, or suggestions that GenAI can automatically format entire manuscripts to fit specific journal guidelines are unfounded. Such formatting usually requires manual intervention to ensure adherence to the specific requirements of each journal.
- **Preliminary peer review:** No widely recognized GenAI tools simulate the peer review process against a vast database of published papers to predict common points of contention or interest. While there are AI-based grammar and style checkers that can help improve the readability and quality of writing, the nu-

anced feedback provided by human peer reviewers, which includes evaluating the validity of the research methodology, the interpretation of results, and the overall contribution to the field, cannot be fully replicated by AI at this time.

- **Gauging novelty:** GenAI tools can help identify existing literature on a topic, but their ability to assess the novelty of research is limited. Determining the unique contributions of a study requires deep understanding and contextual knowledge that currently goes beyond the capabilities of AI.
- **Creating a list of suitable reviewers:** Databases and networking platforms can help identify researchers with relevant expertise, but the selection of suitable reviewers typically involves editorial judgment and knowledge of the field. GenAI does not currently manage this aspect of the publication process.

In summary, while AI can provide support in the academic publication process, its role is primarily in assisting with research, organization, and some aspects of writing. The more nuanced tasks, such as peer review, assessing research novelty, and selecting suitable reviewers, still rely heavily on human expertise. It's essential for you to remain critically engaged and not overestimate the capabilities of AI in the context of academic publishing.

Developing White Papers and Reports Using Generative AI

GenAI is helpful in writing white papers and reports, which are often extensive and require a significant investment of time and expertise. There are limitations to how much of the work GenAI tools can perform. Here's an overview of where things stand in terms of GenAI assisting with white papers and reports as of this writing.

- **Automated drafting:** While GenAI can assist in creating outlines and initial drafts, it's important to note that these tools typically require human input to guide the structure and ensure that it meets the specific goals of the white paper or report. The AI's ability to automate drafting is based on patterns it has learned from data, so it might not always align perfectly with the author's intentions without manual adjustments.
- **Suggesting content based on trends:** GenAI does have the capability to analyze large datasets and can identify trends to some extent. However, the insight it provides is often based on the data it has been trained on, and it may not have access to the most current or proprietary databases unless integrated by the

user. Predictive models in AI are also not infallible and should be used with caution, especially when informing strategic decisions.

- **Synthesizing information:** GenAI can indeed process and summarize information from various sources, but the quality and depth of synthesis can vary. It's essential for the user to critically evaluate and supplement AI-generated summaries to ensure that they capture the necessary details and context accurately.
- **Accuracy and relevance:** It is crucial to highlight that verifying the accuracy and relevance of AI-generated content is a key step in the process. GenAI can sometimes replicate misinformation if that's present in the training data, and it may not always interpret data correctly, especially if it involves complex or nuanced topics.



REMEMBER Even though GenAI can support the development of white papers and reports by providing structural outlines and synthesizing information, its capabilities are not a substitute for human expertise and judgment. The technology serves best as an aid in the writing process, with the author playing a critical role in ensuring the final document's accuracy, relevance, and credibility.

Crafting professional documents with AI support

The integration of AI into the process of creating professional documents such as white papers and reports can significantly enhance productivity and quality. AI support can streamline research, organize content, and even aid in the editing process, allowing writers to focus on the critical thinking and strategic aspects of document creation.

When starting a white paper or report, you can use AI tools like Grammarly (`www.grammarly.com`) and Hemingway (`hemingwayapp.com`) to help refine the language and improve readability. These tools provide real-time suggestions to make the text clearer, more concise, and grammatically correct. For more complex editing tasks, tools like ProWritingAid (`prowritingaid.com`) offer in-depth reports on style, word choice, and sentence structure.

For research, AI-powered tools such as Yewno Discover and Connected Papers (`www.connectedpapers.com`) offer innovative ways to explore

the academic landscape. They can help authors find the most relevant studies, papers, and articles by understanding the context of their queries, not just matching keywords. This can lead to a more informed and up-to-date background section in any professional document.

AI can also help in organizing the document's structure. Tools like Scrivener (`www.literatureandlatte.com/scrivener/overview`) help writers manage large writing projects by breaking them down into smaller, more manageable sections. This can be particularly useful for lengthy reports or white papers that cover multiple topics or require extensive data presentation. However, Scrivener does not directly use or provide AI tools. Most people who want to use GenAI tools copy and paste the documents they created in Scrivener to a GenAI tool to further their work.

For data analysis and visualization, tools like Tableau (`www.tableau.-com`) and Microsoft Power BI leverage AI to help users identify trends, patterns, and insights from complex datasets. These can be incorporated into professional documents to support arguments with empirical evidence and create compelling visualizations that convey the data's story at a glance.

Meanwhile, AI-driven summarization tools can condense large volumes of information into concise overviews. For instance, tools like Resoomer (`resoomer.com`) can be useful when one needs to quickly assimilate and report on the key points from a range of sources or when creating an executive summary for a detailed report.



WARNING However, though AI provides fast and intensive support, I'll again remind you that these tools are not infallible. You should always review AI-generated content for accuracy, tone, and relevance. It's also important to ensure that the use of AI aligns with ethical guidelines and intellectual property laws, especially when dealing with proprietary or sensitive information.

Enhancing business communication with AI

This section focuses mostly on white papers and reports, but GenAI can assist with other long-form business communications as well.



TIP Following are some general tips for using GenAI on any long-form business or academic content:

- **Define clear objectives:** Before using AI, define the goals of your document. What message do you want to convey, and to whom? Clear objectives will help you guide the AI in generating relevant content.
- **Validate AI contributions:** Always factcheck and validate AI-generated content, especially when dealing with technical subjects or data analysis.
- **Balance technicality and readability:** Use AI to strike a balance between technical accuracy and readability. Tailor the complexity of the language to the document's intended audience.
- **Keep the human element:** Ensure that the final document reflects the company's voice and values. AI should support the communication strategy, not define it.
- **Stay updated on AI developments:** The field of AI is rapidly evolving. Stay informed about new tools and updates that can further enhance your business communications.

Crafting Research Designs and Outlines with GenAI

GenAI can streamline the research process, from conceptualization to data collection, including crafting research designs, outlines, and poll questions. GenAI tools are adept at organizing thoughts and structuring complex information, making them particularly useful for researchers who are in the early stages of project development.

When it comes to research design, GenAI can assist in developing a comprehensive outline that covers all necessary components of a study. By inputting the research objectives and questions, researchers can use AI to generate a framework that includes the introduction, literature review, methodology, data analysis, and conclusion sections. AI tools like

ZoteroBib (`zbib.org`) can also help researchers manage their sources and citations efficiently, ensuring that their literature reviews are thorough and properly referenced. Moreover, AI-driven platforms such as Iris.ai can explore vast databases to identify relevant studies and theoretical frameworks, suggesting connections that may not be immediately obvious but that you would want to include in the research design.

For creating poll questions and surveys, GenAI can be particularly useful in formulating questions that are clear, unbiased, and likely to yield informative responses. AI tools like SurveyMonkey's Question Bank offer a repository of pre-vetted questions that can be customized to fit the specific needs of a study. These AI-generated questions can serve as a starting point, which researchers can then refine to ensure that the wording is precise and tailored to their target demographic. Additionally, AI can analyze pilot survey data to identify poorly performing questions, allowing researchers to make necessary adjustments before launching a large-scale data collection effort.

It is crucial for researchers to critically evaluate and adapt AI-generated content. The researcher's expertise is essential in interpreting AI suggestions within the context of their specific field and research goals. It is this synergy between human intellect and AI efficiency that can produce well-crafted research designs and tools, ultimately leading to more robust and insightful studies.

Structuring research projects with AI

In general, structuring a research project involves the following actions:

1. Define the research question.
2. Conduct a literature review.
3. Develop a methodology.
4. Create a research proposal.
5. Plan the project.
6. Collect data.
7. Analyze data.
8. Write the report.
9. Revise and edit.
10. Peer review (if applicable).

11. Disseminate findings.
12. Reflect on the process.



REMEMBER AI offers valuable tools for assisting with the structuring of research projects, but its capabilities are currently more supportive than autonomous. AI can provide insights and highlight patterns that may not be immediately apparent to researchers, but the tasks of identifying gaps in literature, developing theoretical frameworks, and designing methodologies still require significant human input and critical evaluation.

Consider the following caveats:

- GenAI can help identify gaps in literature by processing large volumes of data and highlighting areas that lack research. However, the AI's effectiveness in this task depends on the quality of the data it has been trained on and the sophistication of the algorithms used. AI can pinpoint underexplored areas or contradictions, but the technology typically requires human guidance to interpret the significance and relevance of these findings within a specific research context.
- GenAI can analyze patterns and correlations in data, which may suggest potential theories. However, AI cannot reliably develop theoretical frameworks or hypotheses. GenAI can provide data-driven insights that may inspire researchers to formulate new hypotheses, but the creation of a robust theoretical framework remains a complex task that relies heavily on human expertise and critical thinking.
- AI does have predictive analytics capabilities, and these can be used to some extent to inform research design based on historical data and outcomes from similar studies. However, the ability of AI to guide the methodological design of a project is not as straightforward as is often suggested. Predictive analytics can provide some foresight into potential outcomes, but human researchers need to interpret these predictions and make methodological decisions based on a comprehensive understanding of the subject matter.

From concept to outline: GenAI as a planning tool

GenAI can play a significant role in the initial stages of academic and professional research. Its capacity to process and organize information provides a foundational tool for you to transition from a broad concept to a structured research outline.

At the conceptual stage, AI can be utilized to conduct preliminary literature reviews and environmental scans. Tools like Iris.ai facilitate this by allowing researchers to input their research questions or abstracts and then receive a map of related work and concepts. This not only speeds up the literature review process but also ensures a comprehensive understanding of the field, helping to refine the research questions and objectives.

Additionally, AI applications can analyze the sentiment and trends within existing literature, offering insights into prevailing research attitudes and potential biases, which can be instrumental in defining the scope and direction of a new study. However, their use in academic literature reviews is not as widespread as in other fields, such as market analysis or social media monitoring. The application of sentiment analysis to academic literature to uncover biases and trends is an emerging area, and while tools exist, they may not yet be a standard part of the research design process.

After your research concept is solidified, GenAI aids in outlining the research design by suggesting logical structures and sequences for the project. GenAI-powered mind mapping tools, for example, can help visualize the connections between various components of your research project, such as hypotheses, methodologies, and expected outcomes. Examples include NoteGPT, MindMatrix, Mapify, and GitMind.

Furthermore, AI can assist in identifying the most suitable research methods for your project by drawing from databases of past studies, analyzing their success rates, and aligning them with the current research goals. This can add efficiency to the planning phase. However, using GenAI to suggest research methods is a complex task and is not commonly found in mainstream GenAI tools. GenAI systems are in development for specialized fields that can analyze and suggest methodologies, though these are typically more accessible to authorized users or research institutions.



REMEMBER Despite the assistance GenAI can provide, the creativity and critical judgment of the researcher remain crucial in crafting a research de-

sign that is both innovative and scientifically rigorous.

Integrating Citations and References

Integrating citations and references is a critical aspect of academic writing and research, ensuring that proper credit is given for the ideas and findings of others. The process can be tedious and time-consuming and is one that I am happy to relegate to GenAI. You probably are, too.

As GenAI technology continues to evolve, we can expect further advancements that will make the process more intuitive and integrated into the research workflow, allowing scholars to focus more on the substance of their work rather than the technicalities of citation. For now, well, let's just say there's a long way to go yet. Integrating GenAI with other more mature research tools is probably your best bet.

GenAI-assisted reference management

The good news is that citation tools, like Zotero and Mendeley, extract reference details from documents, websites, and other digital sources. These tools can recognize and format citations according to various style guides, such as APA, MLA, or Chicago, with minimal input from you. They also allow you to maintain a searchable library of references, which can be easily organized by topic, project, or other custom tags. This not only saves time but also reduces the risk of errors in citation, which is crucial for maintaining the integrity of a research project.

The bad news is that most, if not all, pure-play citation and reference management tools like Zotero and Mendeley (`www.mendeley.com`) are not GenAI-driven as of this writing. Some will allow you to integrate them with GenAI. There's much debate about whether integrating GenAI will strengthen or weaken these tools in the end.

But it is true that some advanced word-processing programs and writing platforms have begun to more fully incorporate GenAI-driven features. For example, tools like Grammarly and Microsoft Editor use GenAI to help improve writing quality, which can include suggestions for citations.

However, a feature that suggests relevant articles to cite based on the content being written is not commonly found across all word-processing programs and may only be part of more specialized or advanced writing assistance software.

It's reasonable to expect that GenAI technology will continue to evolve and further integrate into the research workflow. But the availability and sophistication of these features can differ significantly depending on the specific software or platform in use. As GenAI becomes more sophisticated, its potential to streamline the research process and allow scholars to focus more on the content of their work is likely to increase.

Ensuring academic integrity with GenAI

Many modern writing assistance tools and plagiarism-detection services, such as Turnitin and Grammarly, use sophisticated algorithms and GenAI to scan texts against databases of academic work to identify potential plagiarism. These tools can result in users adding citations where necessary to help ensure that all sources are properly credited.

Moreover, GenAI can assist in paraphrasing and summarizing existing research while maintaining the original meaning, which helps in avoiding plagiarism. GenAI writing assistants can suggest alternative ways to present information that ensure the researcher's prose is unique, yet still reflects a thorough understanding of the subject matter. This is particularly useful for students and researchers who are synthesizing complex information from multiple sources.

GenAI can also contribute to academic integrity by ensuring consistency and accuracy in data reporting. GenAI algorithms can analyze datasets for anomalies or inconsistencies that may indicate errors in data collection or analysis. By flagging these issues, researchers can address them proactively, ensuring that their reports are as accurate and reliable as possible. As GenAI continues to evolve, its potential to reinforce ethical practices in academic research is likely to expand, providing researchers with a powerful ally in the quest for integrity and excellence in their scholarly endeavors.



WARNING But GenAI tools can and do hallucinate, so the onus remains on you to take every measure beyond GenAI that is necessary to ensure academic integrity in every work produced with or without GenAI assistance.

Producing Long-Form Articles with GenAI

The use of GenAI in producing long-form articles has certainly brought a mixed bag of outcomes — good, bad, and ugly. On the positive side, GenAI has democratized content creation, enabling writers, marketers, and researchers to produce comprehensive articles more efficiently. It can assist with generating ideas, providing structure, and even drafting sections of content, which can be particularly beneficial for those facing writer's block or tight deadlines. Additionally, GenAI can help non-native speakers articulate their thoughts more fluently, broadening the inclusivity of content creation.

However, the use of GenAI in writing long-form articles is not without its pitfalls. The *bad* often emerges in the form of over-reliance on AI, which can lead to homogenized content lacking in depth, personal insight, or the nuanced understanding that human expertise brings. There's also the risk of factual inaccuracies or the perpetuation of biases present in the training data, which can mislead readers and erode trust in the content. Furthermore, the *ugly* side surfaces when GenAI is used unethically to mass-produce deceptive or plagiarized content, manipulate opinions, or spread misinformation, which can have far-reaching negative consequences on public discourse and trust in media.

While GenAI offers powerful tools for streamlining the writing process and enhancing productivity, it also poses significant challenges that need to be managed with a critical eye toward maintaining quality, originality, and ethical standards. As the technology continues to evolve, it will be crucial for writers and publishers to balance the efficiency gains with a commitment to the integrity and value of the written word.

Techniques for GenAI-enhanced feature writing

Feature writing typically allows a little more time to complete than news writing because it usually involves more research and interviews and isn't necessarily tied to the day's headlines. These are the pieces that typically garner many views and are the stuff that magazines and legendary writers and reporters built their reputations on.

Today, feature stories still take a deeper dive into their subject and often offer nuance and color that straight news reporting lacks. The question now becomes whether feature writing is a good use case for GenAI — or not? The answer is it depends.

A lot depends on the talent and experience of the writer and the capabilities of the GenAI model used. I find that *output stitching* — in which you give multiple GenAIs the same prompt and then stitch the outputs together —is a highly effective technique for writing most long-form articles, but most especially features. (Yes, *GenAIs,* plural, meaning more than one GenAI.) Using multiple GenAIs gives me many different options and perspectives that a single model just can't deliver. I can also break repetitive outputs and other annoying GenAI behaviors by using *AI chaining*, which is basically infusing other GenAI outputs in the input (prompt) of another. You'll be amazed at how well that works.



TIP I also find AI aggregation to be a crucial strategy as that enables me to use a variety of GenAI built for different purposes such as background research, financial data analysis, cross-referencing and cross-factchecking, source discovery, image and data visualization generation, and other tasks I might need to bring to bear to create really great long content. And I've done all this testing, strategizing, and process building for projects like this book — to see what works and what doesn't — and to share those findings with you.

If you're producing feature articles for marketing or business purposes outside of journalism, you'll find these same techniques to be invaluable.

Meanwhile, here are some techniques and tips for leveraging GenAI in feature writing:

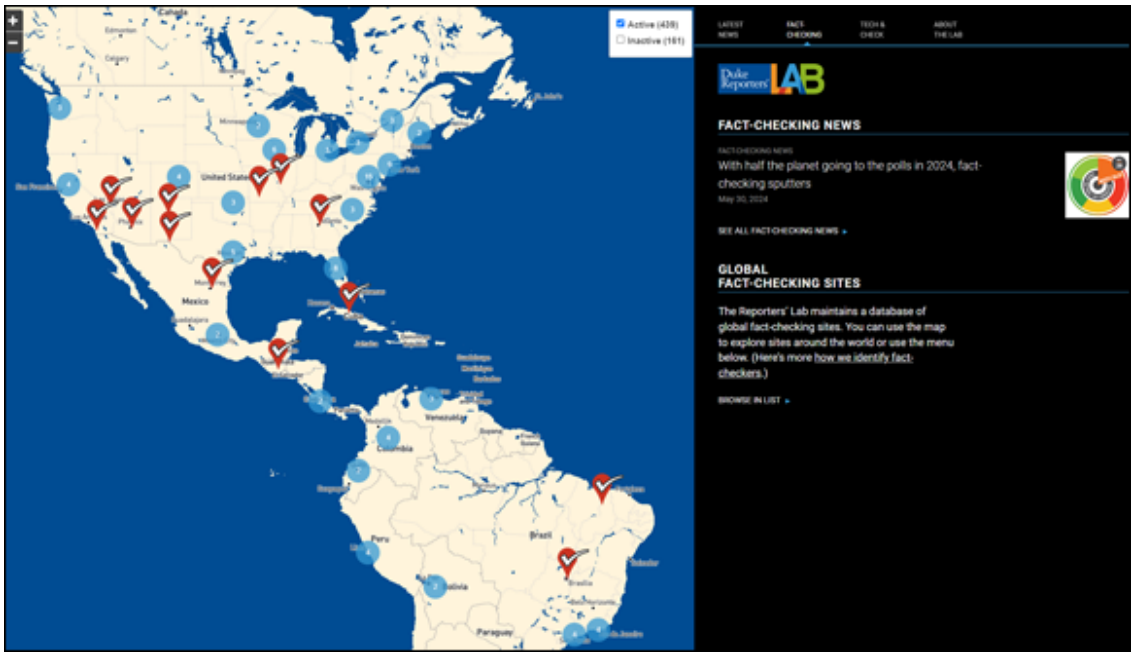
- **Research assistance:** Use GenAI to conduct preliminary research on your topic. Tools like Evernote can help you collect and organize information, while others like ChatGPT and Claude can summarize lengthy reports, contracts, agreements, interview transcriptions, videos, or articles, saving you time and helping you to quickly gather background information and context for your feature.
- **Trend analysis:** GenAI can analyze large datasets to identify trends that might be relevant to your story. Tools like Google Trends or social listening platforms can provide insights into what topics are currently popular or rising in public interest, which can help you to choose angles that will resonate with your audience.
- **Writing assistance:** GenAI writing assistants can help you overcome writer's block by suggesting sentence completions or alternative phrasings. While you should not rely on these tools to write large sections of your article, they can be useful for getting started or finding a new way to express an idea.
- **Editing for clarity and style:** GenAI-powered grammar and style checkers can help refine your writing, ensuring that it is clear, concise, and free of errors. Tools like Grammarly or Hemingway Editor can suggest improvements to sentence structure, word choice, and readability.
- **Audience engagement:** Use AI to understand your audience better. GenAI and GenAI-integrated analytics tools can track reader engagement and provide feedback on which parts of your articles are the most and least engaging. This information can be invaluable in refining your writing style and content to better match your readers' interests.
- **Personalization:** GenAI can help personalize content for different segments of your audience. By analyzing reader data, GenAI can suggest topics, stories, and approaches that are more likely to appeal to specific demographics or reader groups.
- **Factchecking:** Use GenAI factchecking tools to help verify the accuracy of statements and data presented in your feature. This is crucial for maintaining credibility and trust with your readers. You might want to check out Full Fact in the UK at `https://fullfact.org/` and the map and info on many global factchecking sites at Duke Reporters' Lab at `https://reporterslab.org/fact-checking/`, which is pictured in Figure 4-1.



REMEMBER GenAI is a tool to assist you in the writing process, not a replacement for journalistic integrity and ethical considerations. Be transparent about the use of GenAI in your writing process and ensure that the final product reflects your voice and adheres to the highest standards of journalism.

Maintaining depth and quality in GenAI-assisted articles

To preserve depth, you should use GenAI for initial research and drafting but then critically evaluate and expand upon the information provided by GenAI, incorporating your own knowledge, perspectives, and additional research. This helps to add layers of understanding that AI alone cannot achieve. You should also factcheck all GenAI-generated content to ensure accuracy and reliability, as even specialized GenAI can and does hallucinate on the regular as well as perpetuate errors or biases present in its training data.



OpenStreetMap / `https://reporterslab.org/fact-checking` // last accessed on April 23, 2025

FIGURE 4-1: Screenshot of the interactive map of global factchecking sites at Duke Reporters' Lab.

Quality is upheld by using GenAI tools to enhance your natural voice and improve readability without compromising your unique style. GenAI can

assist with grammar, syntax, and structure, but you should always have the final say in how the content is presented. Additionally, you should remain mindful of ethical considerations, giving proper attribution when necessary and being transparent about the use of GenAI in your writing process.

Writing Books with GenAI

If you're prompting GenAI to write complete books or, heaven forbid, thinking of it as a coauthor, you're doing it wrong. That's like an architect sharing the credit with CAD software for the creation of a beautiful building. I mean, I guess you can do that if you want, but you're dissing your own talent and leading others astray in their work by implying that the software can do much more than it does. Instead, just make it clear that you used GenAI to make your own creative work until the day comes around that it's as routine as an architect using CAD in the course of their work.

The most important thing to manage is your own expectations. GenAI is awful at long-form writing and even worse at writing books. Ask the publishers and self-publishing services how many truly awful GenAI-generated books were thrust upon them by people who thought they could make a fortune at the push of a button. A flood of trash came over the transom and went right back out to the trash bin.



REMEMBER The key to writing books with GenAI is to take the lead. Don't let GenAI models lead you. They are not omnipotent. And they lie just as easily as any other source you might interview. Remember that. Don't trust and always verify.

Decide what your book or story is about and use GenAI to help you research, organize, and write it one small chunk at a time. Yes, you're going to have to make GenAI spit out the proverbial elephant one bite at a time and then stitch it all together yourself. But first, you're going to have to describe and refine and prompt and re-prompt and *prompt chain* every

one of those chunks so that you don't stitch together some other animal parts that are distinctly not of the elephant species.



TIP Pay attention to output stitching, AI aggregation, and AI chaining as you'll likely need to use all of them when creating any long-form content, especially in writing a book, regardless of the topic, style, or format. You'll also need to make yourself adept at prompt chaining, a technique in which a sequence of interconnected prompts is used to guide a GenAI model through a multi-step reasoning process or to build upon previous outputs for more complex tasks. In other words, prompt chaining in this context is prompting one chunk at a time so you don't confuse or distract the GenAI by giving it too much information or too many tasks at a time.

How and when you use each of these techniques is up to you. This works in much of the same way as you deciding which words to use and in what order to write them to convey your meaning or story to your readers. Sounds like an overwhelming task, and sometimes it is, which is why writer's block is a real thing. But other times, you find yourself in the zone, and words flow as easily as a river. Using GenAI to help you write a book will eventually flow like that for you, too. With time and practice, you will assimilate it into your personal writing process.

Meanwhile, you may also find some of the following helpful, depending on your personal writing style and process and the genre of your book:

- **Outline and structure:** Tools like Plottr (`plottr.com`) or Scrivener can help with outlining, plotting, and structuring your book. They can also be used in conjunction with GenAI suggestions from tools like Sudowrite (`https://sudowrite.com`) or ShortlyAI (`www.shortlyai.com`) to write or plot your narrative.
- **Character development:** Reedsy's Character Builder is a tool you might want to consider when creating detailed character names and profiles. For GenAI-generated backstories and traits, you could use tools like CharacTour (`www.charactour.com/hub`) or AI Dungeon (`aidungeon.com`) for inspiration. Many more general-purpose and multimodal GenAI models can help you develop characters on their own.

- **Drafting assistance:** OpenAI's GPT-3, accessible through platforms like Jasper (formerly Jarvis) or Rytr, can help with generating text for different sections of your book. ChatGPT 4, ChatGPT 4o, Claude, Scribble Studio, and others are helpful, too. (Repeat after me: chunk writing!) These can be especially helpful for drafting dialogue or descriptive passages.
- **Research:** For summarizing information and conducting research, you might use GenAI tools like Socratic by Google, which can process and analyze large amounts of data quickly. You might also want to use Perplexity.AI as your search engine as it typically renders great results and cites its sources too, which makes factchecking its responses much easier. You can also prompt it to deliver even more sources. Also, look at the academic writing section in this chapter for a list of more research tools.
- **Language and style:** Grammarly and ProWritingAid offer GenAI-powered suggestions for language and style improvements, helping to ensure consistency and polish in your writing. But for the most part, you'll find most GenAI text and multimodal generators perfectly capable of doing these same tasks without the need for additional tools.
- **Editing and proofreading:** Hemingway Editor can help tighten up prose, while Grammarly can catch more nuanced grammatical issues and offer stylistic suggestions. But again, you may find more generalized GenAI tools to be sufficient in these tasks without the need for specialized tools.
- **Feedback and revision:** Although not a replacement for human feedback, platforms like Authors.ai (`authors.ai`) provide GenAI manuscript analysis, offering insights into how your book might resonate with readers based on existing successful novels.
- **Ethical considerations:** No specific GenAI tool is designed to manage ethical considerations, as this is more about how you approach the writing process. However, tools like Grammarly and Turnitin can help ensure originality and proper citation in academic and non-fiction writing to help maintain ethical standards.



REMEMBER The quality of your book will ultimately depend on your skill as a writer and your ability to effectively incorporate GenAI-generated content into a cohesive and engaging narrative. In short, great book writing is a credit to the writer, not the tools.